ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 2 · PRACTICE D

MARKS: 80 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 2. Laid out like the DHET second paper (Fritzing practical, 80 marks): an LED-and-buzzer circuit built from a description, a transistor NOT gate and a Darlington rain alarm built on a breadboard, and an LED fade-out circuit drawn from a breadboard layout.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of TWO questions. Answer ALL the questions.
2. This is a practical paper. Use the Fritzing software for all circuits and breadboard layouts. If you do not have a computer, draw the circuits and layouts neatly by hand.
3. Put your name or examination number on every printout or page, and place the printouts in your examination folder.
4. Show the positive and negative rails clearly and make sure every component has the correct polarity.

QUESTION 1

1.1 Use Fritzing to draw a circuit with the following specifications: (20)

- Three red LEDs and ONE 220 Ω resistor are connected in series.
- A buzzer is connected in parallel with this LED string.
- A slide switch, in series with the supply, switches the whole circuit on and off.
- The circuit is supplied by a 9 V battery.

1.2 Build the circuit diagram below on a breadboard using Fritzing. The transistor works as a NOT gate: LED1 shows the input and LED2 shows the output. (20)

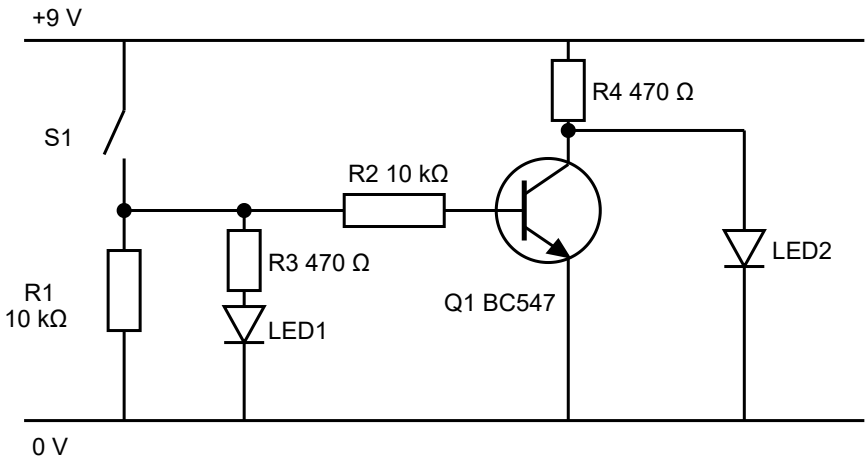


FIGURE 1: Transistor NOT gate

[40]

QUESTION 2

2.1 Build the circuit diagram below on a breadboard using Fritzing. When water touches both probes, the Darlington pair switches on the buzzer and the LED. (20)

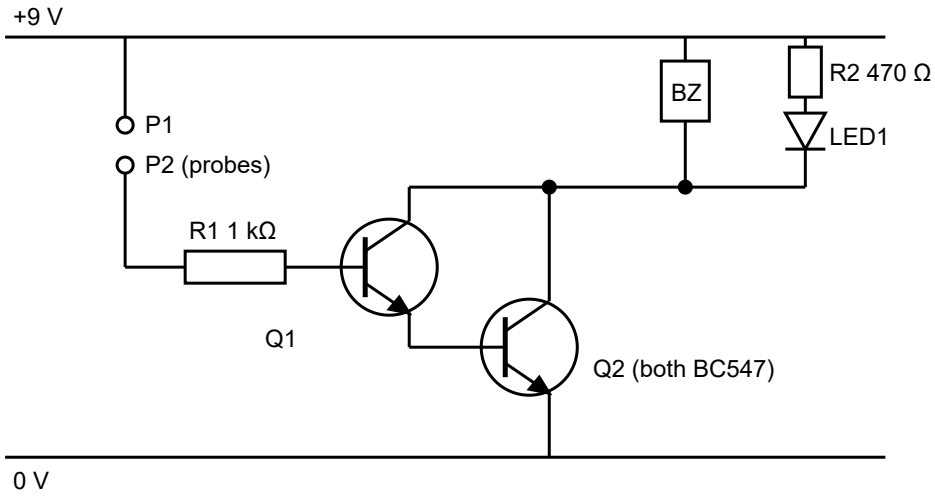


FIGURE 2: Darlington rain alarm

2.2 A learner built the circuit described in the table below on a breadboard. Use Fritzing (or draw by hand) to draw the CIRCUIT DIAGRAM of this layout. When the button is pressed and released, the LED fades out slowly. (20)

Component	Where it is connected on the breadboard
9 V battery	+ lead to the top (+) rail; − lead to the bottom (−) rail
Push button S1	one pin in the + rail, the other pin in row 4
1 kΩ resistor R1	from row 4 to row 8
470 µF capacitor C1	+ lead in row 8, − lead in the − rail
1 kΩ resistor R2	from row 8 to row 12
Green LED	anode (long leg) in row 12, cathode in the − rail

[40]

TOTAL: 80

END OF QUESTION PAPER